

Document Date: 14 August 2026

Infosys Springboard Virtual Internship 7.0 Completion Report

Team Details

Batch Number: 1

Start Date: 29 June 2026

Members Names:

Himadri Ketan Tripathy, Asheesh Patel, Likhita Perakam, Govardhanachari Vadla, Harshita Akula, J Gopi Sai, K. BhagyaSri Krishna Sai Akhshaya, M Yaswanth Rakesh, Noudu Sai Durga Jyothsna, Dhruv Raj Singh, Harshita Bolisetty, Harini Jeyashree A, Disha B G, DhanusKodi, Papaiahgari Madhuri Reddy, Palivela Kusuma Nageswari, Gangisetty Halini Rama Madhuri, Syed Sameer, Josna Fernandes, Akshata Laad.

Internship Duration: 8 Weeks

1. Project Title

AI-Based Nutritional Assessment System for Deficiency Detection and Personalized Meal Planning

2. Project Objective

The objective of the project was to develop an AI-powered nutritional assessment platform that can identify nutritional deficiency risks, support personalised meal planning, and help users monitor dietary habits. The system brings together machine-learning-based deficiency prediction, health and symptom assessment, blood-report processing, food information, dietary tracking, and dashboard-based insights within a single application.

3. Project description in detail

Core modules: deficiency-risk prediction using Random Forest and XGBoost; personalised meal planning; food diary; blood-report processing; symptom assessment; food scanner and barcode lookup; dashboard; AI Coach; multilingual support; PWA functionality; admin panel; JWT authentication; Google Gemini 2.0 and Open Food Facts integrations.

The latest implementation uses a React 18 with Vite PWA frontend and a FastAPI backend connected to PostgreSQL through RESTful APIs secured with JWT authentication. The machine learning layer uses Random Forest and XGBoost for nutritional deficiency risk prediction. Google Gemini 2.0 supports AI-assisted nutrition estimation and the AI Coach, while Open Food Facts provides barcode-based product and nutrition information. The system includes a dashboard, food diary, meal planning, risk-score presentation, food scanner, multilingual support, PWA functionality and an admin panel. The frontend is deployed through Vercel and the backend through Render. My individual contribution

focused on the ML/application integration workflow, testing and validation, and project coordination as Team Lead.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1 29 Jun–05 Jul	Kick-off; understand requirements, architecture and AI/ML workflow.	Kick-off, project explanation and setup guidance completed; Milestone 1 responsibilities established.
Week 2 06–12 Jul	Milestone 1 implementation; technical/status guidance; GitHub KT.	Initial implementation progressed; Milestone 1 submitted on 12 July 2026.
Week 3 13–19 Jul	Milestone 2 planning; application and ML integration.	Milestone 2 implementation progressed through technical guidance.
Week 4 20–26 Jul	Milestone 2 refinement; service and integration validation.	Milestone 2 reviewed and submitted on 26 July 2026.
Week 5 27 Jul–02 Aug	Milestone 3 planning; frontend, backend and ML/AI integration.	Milestone 3 integration and technical validation progressed.
Week 6 03–09 Aug	Milestone 3 testing, evaluation and final validation.	System-wide validation completed; Milestone 3 submitted on 9 August 2026.
Week 7 10–16 Aug	Technical completion; final review; documentation/PPT and presentation preparation.	
Week 8 17–21 Aug	Final PPT/document submission, presentation/demo mocks and project closure	

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kick-off	Kick Off Call, AI usage introduction, project discussion and setup guidance.	29 June 2026
Prototype/First Draft	Milestone 1 submission following initial project setup and implementation.	12 July 2026
Mid-Term Review	Milestone 2 submission following implementation, technical guidance and status reviews.	26 July 2026
Final Submission	Individual and Team Document Submission after technical completion and final review.	14 August 2026
Presentation	Mock 1 of Presentation and Demo; further mocks scheduled thereafter.	

5b. Project execution details

The AI-Based Nutritional Assessment System for Deficiency Detection and Personalized Meal Planning is a full-stack AI-enabled web application designed to provide personalised nutritional assessment and dietary support. The major components and functionalities of the project are:

1. Frontend Development

- o Developed using React 18 with Vite and implemented as a Progressive Web Application (PWA). o Provides user-friendly interfaces for the dashboard, assessment, food diary, meal planning, prediction results and food scanning.
 - o Includes responsive UI, interactive components, loading/error states and multilingual interface support.
2. Backend Development o Implemented using FastAPI with RESTful APIs. o Handles authentication, user profiles, dashboard processing, food diary, meal planning, assessment and AI/ML services.
- o Uses JWT-based authentication for protected user operations and role-based access for administrative functionality.
3. Database Management
- o Uses PostgreSQL for persistent application data storage. o Stores information related to user profiles, health data, predictions, food diary records, meal plans and progress information. o Database integration was validated through end-to-end application testing.
4. Machine Learning Integration
- o Includes a dedicated machine-learning layer for nutritional deficiency-risk prediction.
 - o Random Forest and XGBoost models are used for deficiency-risk prediction. o The ML workflow includes preprocessing, model inference, risk-score generation and presentation of prediction results through the application.
5. AI-Assisted Nutrition Services o Google Gemini 2.0 is integrated for AI-assisted nutrition-related functionality. o Supports nutrition estimation and the AI Coach functionality. o The system processes AI-generated responses and presents the results through the application interface.
6. Food and Nutrition Integration
- o Open Food Facts API is integrated for barcode-based product and nutrition information retrieval. o The food-scanner workflow allows product information to be retrieved and displayed through the application. o External service integration was tested as part of the end-to-end validation.
7. Personalised Nutrition Features o Provides nutritional assessment and deficiency-risk information.

- o Includes a personalised meal-planning workflow.
- o Provides food diary functionality for recording dietary information.
- o Dashboard components present nutritional and progress-related information to users.

8. System Integration and Testing

- o The complete workflow was tested across the frontend, backend, REST APIs, database, ML/AI services and external APIs.
- o Frontend–backend communication, authentication, database persistence, prediction workflows and service integrations were validated.
- o UI/UX, responsive behaviour, loading/error states and major user workflows were also reviewed.

9. Deployment and Validation

- o The frontend is deployed through Vercel, while the backend is deployed through Render.
- o The deployment workflow and health-check mechanisms were validated.
- o A Render health-check issue related to HTTP HEAD requests was identified and resolved by updating the backend routing, after which the health checker returned HTTP 200 OK.

10. Overall Impact

- The project integrates Artificial Intelligence, Machine Learning, web development, database management and external data services into a single application.
- It provides an integrated platform for nutritional assessment, deficiency-risk prediction, personalised dietary planning and nutrition-related assistance.

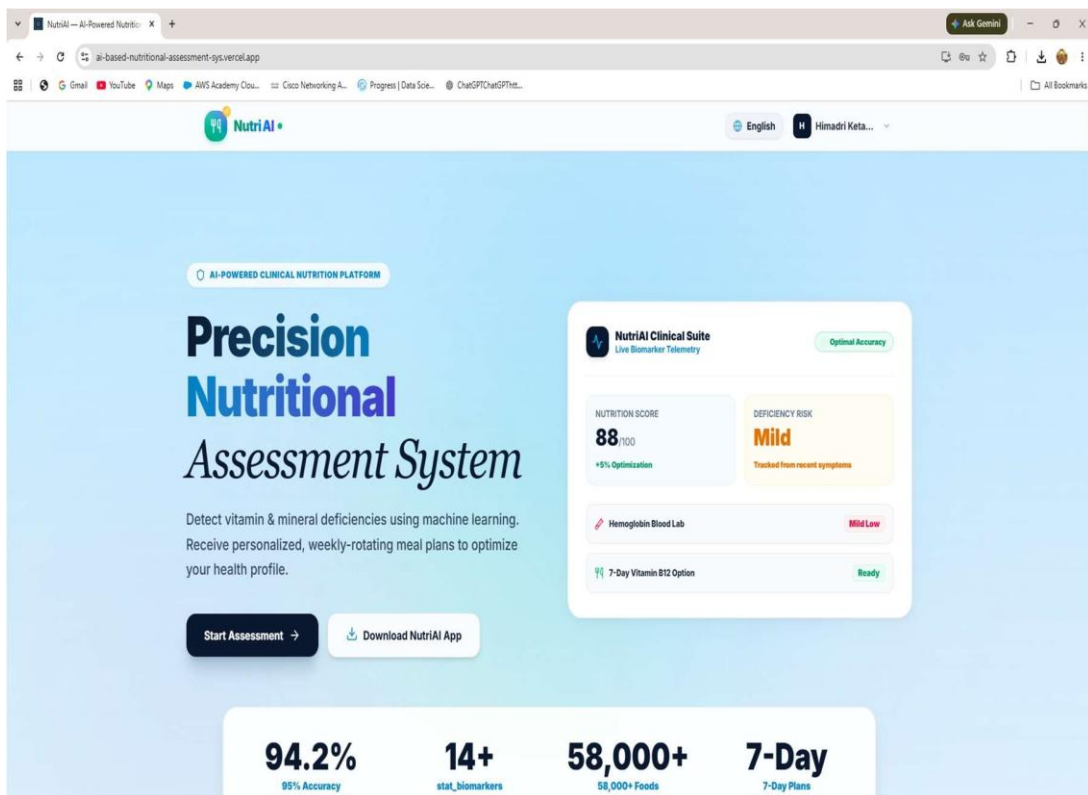
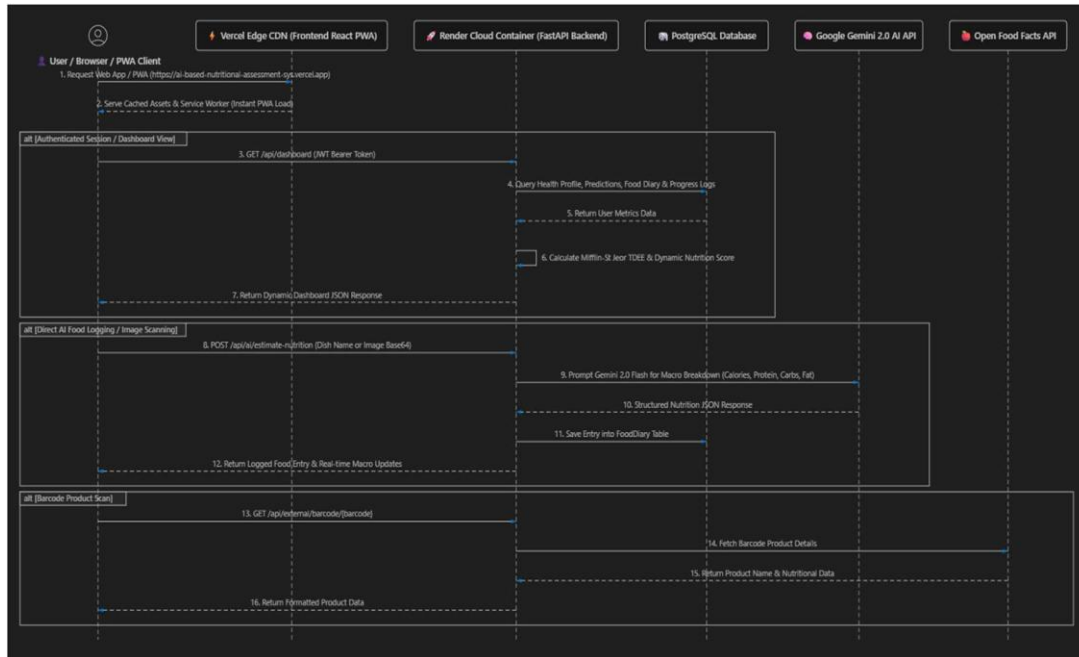
6. Snapshots / Screenshots

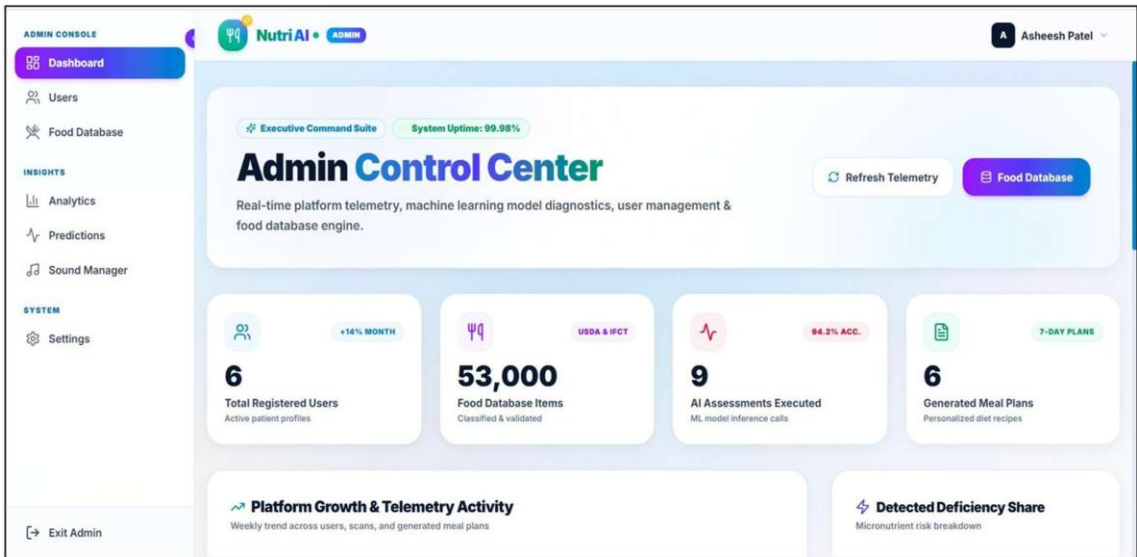
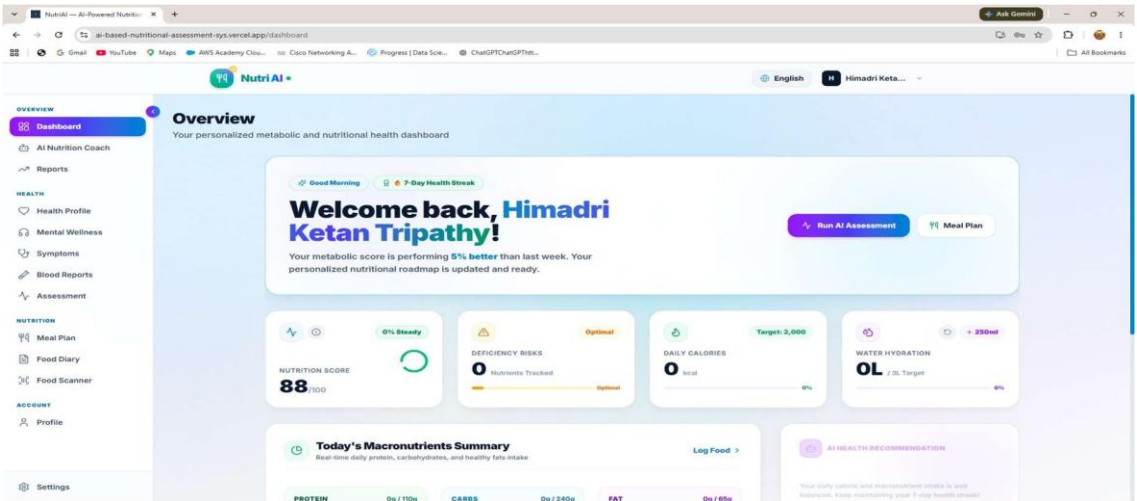
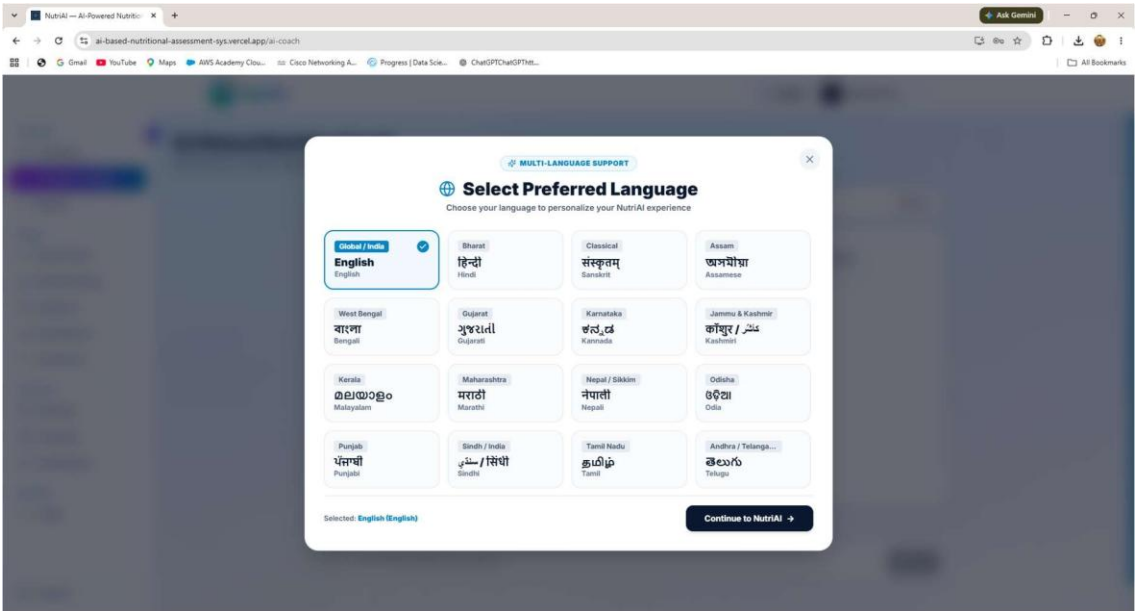
*** MODEL COMPARISON ***

	Target	Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
0	Iron_Deficiency	Random Forest	89.98	88.87	90.84	89.84	96.43
1	Iron_Anemia_Deficiency	Random Forest	93.02	75.16	79.14	77.10	96.65
2	Calcium_Deficiency	Random Forest	87.96	92.06	89.38	90.70	94.86
3	Magnesium_Deficiency	Random Forest	90.27	95.27	92.62	93.92	95.30
4	Potassium_Deficiency	Random Forest	92.78	97.29	94.06	95.65	97.06
5	Iron_Deficiency	XGBoost	93.42	92.40	94.26	93.32	98.52
6	Iron_Anemia_Deficiency	XGBoost	99.21	96.73	98.01	97.37	99.94
7	Calcium_Deficiency	XGBoost	91.11	92.88	93.64	93.26	97.37
8	Magnesium_Deficiency	XGBoost	94.55	96.61	96.67	96.64	97.81
9	Potassium_Deficiency	XGBoost	95.23	96.99	97.38	97.18	98.24

```

File Edit Selection View Go Run Terminal Help
C:\Users\Asus\Desktop> schema.sql
1 |
2 | -- AT-Based Nutritional Assessment System
3 | -- PostgreSQL Database Schema
4 |
5 |
6 | CREATE DATABASE ai_nutrition_db;
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9 | -- USERS
10 |
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12 | CREATE TABLE users (
13 |   user_id SERIAL PRIMARY KEY,
14 |   full_name VARCHAR(100) NOT NULL,
15 |   email VARCHAR(100) UNIQUE NOT NULL,
16 |   password_hash TEXT NOT NULL,
17 |   phone VARCHAR(20),
18 |   role VARCHAR(20) DEFAULT 'USER',
19 |   is_active BOOLEAN DEFAULT TRUE,
20 |   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
21 |   updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
22 | );
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24 |
25 | -- USER HEALTH PROFILE
26 |
27 |
28 | CREATE TABLE health_profiles (
29 |   profile_id SERIAL PRIMARY KEY,
30 |   user_id INTEGER UNIQUE REFERENCES users(user_id) ON DELETE CASCADE,
31 |
32 |   age INTEGER,
33 |   gender VARCHAR(20),
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7. Challenges Faced

During final system validation, an important deployment issue was identified with the Render health checker. The relevant backend endpoint did not initially support the HTTP HEAD request used by the health check, which resulted in a 405 Method Not Allowed response. The backend routing was updated to support both GET and HEAD requests for the required endpoint, after which the Render health checker returned HTTP 200 OK. The project also required systematic validation of frontend-backend integration, API response handling, PostgreSQL persistence, ML/AI service communication, external API workflows, authentication, UI states, multilingual support and PWA behaviour. These issues were addressed through structured testing, integration checks and final verification.

8. Learnings & Skills Acquired

Key learning areas include full-stack application development with React and FastAPI; REST/JSON API integration; SQL Alchemy and PostgreSQL database handling; JWT authentication and role-based access control; machine-learning model integration using Random Forest and XGBoost; Google Gemini 2.0 and Open Food Facts API integration; PWA development and validation; cloud deployment using Vercel and Render; system testing and end-to-end integration; and project coordination and presentation skills.

9. Testimonials from team & Personal Key Contributions

Through the internship, I gained practical experience in machine-learning model integration, Random Forest and XGBoost inference, FastAPI REST APIs, React/Vite application development, PostgreSQL database integration, JWT authentication, external API integration, Google Gemini 2.0 services, Open Food Facts integration, PWA validation, cloud deployment and end-to-end system testing. I also strengthened my communication, task-coordination, presentation and project-management skills through my responsibilities as Team Lead.

My key contribution was supporting the machine-learning and application-integration workflow, particularly the deficiency-risk prediction process and its validation within the complete application. I also participated in system testing and final integration activities across the frontend, backend, database, ML/AI services and deployment workflow. As Team Lead, I coordinated task distribution, followed up on milestone progress, supported communication among team members, and helped keep the project aligned with the internship schedule. The project gave me practical experience in taking an AI/ML component from model integration through application-level validation.

10. Conclusion

The Infosys Springboard Virtual Internship 7.0 provided me with valuable practical experience in applying Artificial Intelligence and Data Science concepts to a real-world nutritional assessment system. My work on machine-learning integration, application validation and system testing helped me understand how AI components can be incorporated into a complete full-stack solution. My responsibilities as Team Lead also strengthened my communication, coordination and project-management skills. Overall, the internship has

improved my technical confidence and provided meaningful practical exposure aligned with my career interests in Data Science, Artificial Intelligence and Machine Learning.

11. Acknowledgements

I sincerely thank Infosys Springboard for providing the Virtual Internship 7.0 opportunity and a project-based learning environment. I am grateful to the mentor, coordinators and my team members for their guidance, feedback and cooperation throughout the project. The technical sessions, milestone reviews, project validation activities and presentation preparation helped me strengthen both my technical and professional skills.